

**Off-Campus Centre of Nitte (Deemed to be University)**

Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

IS3242-1 – GRAPHICS AND ANIMATION

(For IS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. A graphics system typically does not consist of which of the following components?
A) Input devices
B) Output devices
C) Software
D) Pen drive
2. The OpenGL API is primarily used for:
A) Database management
B) Graphics rendering
C) Network communication
D) Text processing
3. Which type of image is created through a process of rendering rather than capturing?
A) Physical image
B) Synthetic image
C) Raster image
D) Vector image
4. Which of the following is NOT a performance characteristic of graphics systems?
A) Resolution
B) Frame rate
C) Latency
D) Network Bandwidth
5. The Sierpinski gasket is an example of:
A) A 2D fractal
B) A 3D object
C) A color model
D) A lighting technique
6. The concept of a "frame rate" is important for:
A) Image quality
B) Rendering speed
C) User interaction
D) Texture display
7. Which of the following is NOT a common application of antialiasing?
A) Image processing
B) Video game graphics
C) Web design
D) Flood fill
8. Control functions in OpenGL are used to:
A) Manage memory
B) Control rendering states
C) Handle user input
D) Define data structures
9. Display lists in OpenGL are used for:
A) Storing textures
B) Organizing data
C) Precompiled graphics commands
D) Handling user input
10. Which of the following is NOT an affine transformation?
A) Reflection
B) Rotation
C) Perspective projection
D) Scaling

11. Picking in graphics refers to:
 A) Selecting objects in a scene
 B) Choosing colors
 C) Editing images
 D) Selecting text
12. The scan-line method is a technique for:
 A) Rendering textures
 B) Visible-surface determination
 C) Color interpolation
 D) Raster till
13. The parametric representation of a parabola can be expressed as:
 A) $y = ax^2 + b$
 B) $y = mx + b$
 C) $x = ay^2 + b$
 D) $J = mx^3 + nx + b$
14. Parallel-projection matrices are used for:
 A) Simulating depth
 B) Maintaining object proportions
 C) Creating realistic shadows
 D) Giving lighting
15. Perspective-projection matrices simulate:
 A) A flat view
 B) Depth perception
 C) Color gradients
 D) Texture mapping
16. Which of the following curves is defined by control points?
 A) Bezier curves
 B) Linear curves
 C) Constant curves
 D) Parabola
17. Key framing in animation helps to
 A) Adjusting lighting effects
 B) Create fluid motion
 C) Creating smooth transitions
 D) Defining important frames in a sequence
18. Global illumination refers to:
 A) Light from a single source
 B) The overall light in a scene
 C) Light reflection and scattering
 D) Light falling on an object
19. Histogram equalization works by:
 A) Redistributing pixel intensities
 B) Adjusting brightness levels
 C) Drawing line
 D) Both (A) & (B)
20. Polygonal shading refers to:
 A) Applying color to polygons
 B) Calculating light reflection on polygons
 C) Both A and B
 D) Apply texture to polygons

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) List and explain any four applications of the Graphics system.	8	L*2	1	1
	b) Explain with equations the object as seen from a pinhole camera.	4	L2	1	1
	c) Write short notes on Display processor and display list.	4	L2	1	1
2.	a) Write the GL program for drawing a smiling face(reshape,keyboard and idle functions not needed).	8	L3	2	2
	b) Distinguish between immediate and retain mode of operation in display's.	4	L2	2	1
	c) Explain orthographic projections in OpenGL with gluOrtho2d() and an example.	4	L3	2	1
3.	a) Write the four major tasks in Geometry processing and rasterization(clipping, projection, rasterization).	8	L2	2	1
	b) Write short notes on i)Menus ii) Picking.	8	L2	2	1
4.	a) Describe Scalars, points and vectors along with all operations on them viz. scalar-vector multiplication, point-vector addition, vector vector addition, inverse vector.	8	L2	3	1

Unit – II

b)	Describe concatenation of transformation with an example viz. rotation about a fixed point.	8	L2	3	2
5. a)	Describe perspective projection with neat diagram and examples.	8	L2	3	1
b)	Explain Projections and Shadows with neat sketches.	8	L2	3	1
6. a)	Explain Scan line method for visible surface detection with neat diagrams.	8	L2	3	1
b)	Describe how to draw a bezier curve using three points along with linear and quadratic equations.	8	L2	3	1
Unit – III					
7. a)	Explain the Phong lighting model.	8	L2	4	1
b)	Explain with an example Cohen Sutherland line clipping.	8	L2	4	1
8. a)	Describe key framing, deformations and character animations.	8	L2	5	1
b)	Describe the image compression standard- JPEG.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
